

THE RIEMANN ZERO LATTICE AS A WARM CRYSTAL: DIFFRACTION, SCREENING, AND THE TWO-BRANCH RESPONSE

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ABSTRACT. Three companion notes measured how the local statistics of Riemann zeros respond to the prime waves of the explicit formula, and identified a sampling-grid systematic: the gap midpoints on which all such measurements stand are themselves displaced by the primes. Here we turn the grid into the object. Its structure factor $\hat{G}(\omega) = \langle e^{i\omega t_n} \rangle$ — the diffraction pattern of the zero lattice — decomposes into three measured components: *arithmetic spots* at $\omega = \log p^k$, *Bragg combs* damped by the lattice temperature, and a *dark field* lying a factor ~ 25 below shot noise (hyperuniformity seen directly in diffraction). The spots obey an absolute, parameter-free law derived from the explicit formula, $|\hat{G}(\log q)| = \Lambda(q) (L\sqrt{q})^{-1} \cos(\pi\tau) \cdot \text{DW}$, verified to 0.1–4% across fourteen prime and prime-power lines, with phases consistent with pure real (deviations $\leq 0.65^\circ$); the comb brightness tracks the measured window temperature (e^{-c} , ratio 0.885 constant across windows, most of the deviation from Gaussian explained by the measured negative excess kurtosis); and binned over the spots, the pattern rebuilds the Montgomery ramp in the first line-bearing bin — Berry’s semiclassical decomposition observed directly. The grid systematic of the companion revisions closes into an identity: all Gram couplings between wave columns are values of $\hat{G}$, and the regression-dressing lives in the thermal diffuse scattering between the dark field and the comb. On the gap side, the same displacement field yields a derived sum rule for the zero-displacement channel — onset $v = 2\tau$ and saturation scale $2/\pi$ — with Λ -weighting confirmed for prime powers ($k \cdot v(p^k)$ lands on the prime curve to 1–5%); the residual is a *screening function* $D(\tau)$, isolated by injection calibration. Finally, the conditional gap response splits into two branches: to its own spontaneous fluctuations the gas responds like CUE for $\tau \gtrsim 0.3$ (within 3%, reaching 0.3% at the Nyquist wavevector — an agreement test not reducible to pair-correlation statistics), while below the first prime line it is effectively frozen ($D \approx 0.04$); the coherent prime branch instead starts near unity and is screened to 0.48, the two branches crossing at $\tau^* \approx 0.14$. Two candidate mechanisms for the measured prime-power weight deficit are refuted — one by a phase-purity test, one quantitatively — and the surviving constraints are stated as open problems.

1. INTRODUCTION

The first three notes of this program [5, 6, 7] measured the joint law of consecutive-zero gaps δ_n and between-zero maxima of Hardy’s Z , dissected it into explicit-formula channels — an amplitude transmission $w(\tau)$ and a zero-displacement response $v(\tau)$, $\tau = \log p/L$, $L = \log \frac{t}{2\pi}$ — and organized the channels into a response theory: a two-channel linear constraint, a sign crossing at $\tau_0 = 0.447$, a thermal Bragg mirror at the Riemann–Siegel fold that reproduces it, and a mirror (functional-equation) interference channel.

The revision of those notes was forced by a discovery about the *measurement* itself: every per-gap regression stands on the gap midpoints, and the midpoints are not a passive grid — the zeros are displaced by the prime waves, so the grid carries the primes, restricted regression bases transfer omitted-line content into retained coefficients, and a ground-truth test showed

the transfer reaches 0.14–0.24 in absolute w units. The companions correct for this by the complete-basis convention. The present note takes the opposite attitude: *the bending ruler is not a nuisance — it is the cleanest window into the zero lattice that this program has found.* Throughout, the style of the companions is kept: no closed forms are fitted; every ingredient is measured or derived; every claim is accompanied by the test that tried to kill it, and two failed derivations are reported as such.

Data: the six primary amplitude windows of [6] (4×10^4 gaps each, $L = 9.86$ to 12.45), with the auxiliary windows of the companions where noted. All measurements are per-window and pooled; placebo and surrogate controls accompany each new observable.

2. THE DIFFRACTION PATTERN

Define the structure factor of the midpoint grid,

$$\hat{G}(\omega) = \frac{1}{n} \sum_n e^{i\omega t_n}, \quad t_n = \text{gap midpoints},$$

— operationally, the diffraction pattern of the zero lattice. Three components appear (Figure 1).

Arithmetic spots. At $\omega = \log q$, $q = p^k$, the pattern shows sharp spots. To first order in the displacement field u (defined by $t_n = t_n^{\text{smooth}} + u_n$), $\hat{G}(\omega) \approx i\omega \langle e^{i\omega t^s} u \rangle$, so a u -line of amplitude U_q at $\log q$ produces a spot $|\hat{G}| = \frac{\omega}{2} U_q \cos(\pi\tau)$, the $\cos(\pi\tau)$ being the midpoint-averaging factor over one lattice spacing (measured independently by injection, below). The spots are *phase-locked*: across eleven primes, six windows, and eight prime powers, $\arg \hat{G}(\log q) = 180.0^\circ$ – 180.6° , deviations well inside the statistical phase uncertainties (3 – 5° on the strong spots, up to $\pm 24^\circ$ on the weakest), with quadrature parts consistent with zero at the 10^{-4} level — the displacement field is sine-locked, exactly as a real gap-channel response demands.

Bragg comb. In the unfolded frame the lattice’s own reflection sits at $\omega_x = 2\pi k$. Its first-order brightness is the characteristic function of the midpoint phase; the Gaussian (Debye–Waller) estimate e^{-c} , with $c = (2\pi\sigma_u)^2/2$ the measured jitter of the companion notes, overshoots by a constant factor: measured/Gaussian = 0.88–0.89 across all six windows. The measured excess kurtosis of the wrapped midpoint phase (-0.75 to -0.79) explains most of the gap through the Edgeworth correction ($0.32 \rightarrow 0.27$ vs measured 0.256 at $L = 9.86$; the 0.885 ratio is quoted against the unfolded-jitter Gaussian e^{-c} , the Edgeworth chain against the wrapped-phase Gaussian — two baselines differing by convention); a $\sim 6\%$ remainder is left to higher cumulants. The second-order comb is extinguished (measured at the noise floor), as strong dephasing predicts. The comb brightness thus tracks the window temperature window-by-window — a third, independent thermometer for the thermal Bragg model of [7].

Dark field. Off the lines and away from the comb, the pattern is *dark*: the median $|\hat{G}|$ over 978 off-line frequencies (of 1200 scanned) is 2×10^{-4} , a factor ~ 25 below the shot-noise level $1/\sqrt{n}$, and below a shuffled-gap surrogate (which destroys the anticorrelations and raises the floor to 1.4×10^{-3}). Hyperuniformity of the zero gas — number-variance rigidity — is here seen directly as darkness in a diffraction pattern. All arithmetic content is concentrated in the discrete spots.

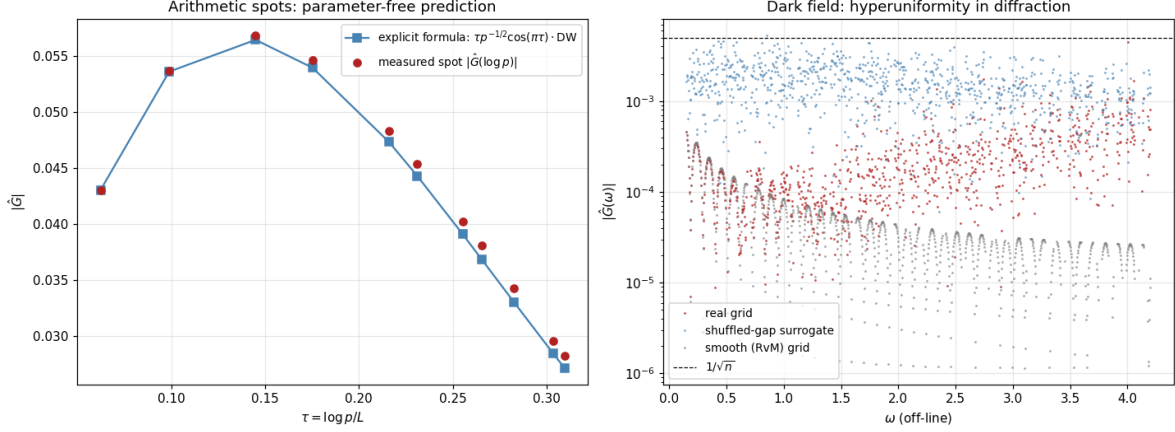


FIGURE 1. Left: arithmetic spots against the absolute explicit-formula prediction of Section 3. Right: the off-line dark field for the real grid, a shuffled-gap surrogate, and the smooth grid; dashed line is shot noise.

3. THE ABSOLUTE SPOT LAW

Derivation 1. *The truncated explicit formula gives the argument function the line content $S(t) \supset -\frac{1}{\pi} \frac{\Lambda(q)}{\sqrt{q} \log q} \sin(t \log q)$, and the zero displacement is $u = -S/\bar{\rho}$ with $\bar{\rho} = L/2\pi$; hence the u -line amplitude*

$$U_q = \frac{2\Lambda(q)}{L\sqrt{q} \log q},$$

and the spot law, with no reference to any channel measurement,

$$|\hat{G}(\log q)| = \frac{\Lambda(q)}{L\sqrt{q}} \cos(\pi\tau) e^{-\omega^2 \sigma_t^2/2}.$$

Observation 2 (Absolute spot law). *Across the six windows, measured/predicted = 0.999 ($p = 2$), 1.002 ($p = 3$), 1.007, 1.012, 1.021, 1.024, 1.030, 1.032, 1.036, 1.039, 1.040 ($p = 31$); mean 1.022 ± 0.014 . The prediction has no free parameters and no channel input: the explicit formula predicts the diffraction pattern of the zero lattice at the percent level. The mild +2–4% rise with ω is consistent with the Gaussian DW factor slightly over-damping a sub-Gaussian field and is left as a second-order refinement.*

For prime powers the same law carries $\Lambda(p^k) = \log p$: the $1/k$ suppression relative to a $\log q$ weighting. Both channels confirm it (Section 5); the strong power spots ($q = 4, 8, 9$) obey the absolute law to 1–4%, while the weak ones fall progressively below it — the *power deficit* of Section 7.

The spot brightness also closes a loop with the gap channel: the ratio of the measured spot to the naive v -based prediction is a pure sampling-geometry factor, $\pi\tau \cot(\pi\tau)$ (the gap regression measures the wave's *difference*, $2\sin(\pi\tau)$, the midpoints carry its *average*, $\cos(\pi\tau)$), times a constant: measured ratio/ $f(\tau) = 1.083 \pm 0.007$, flat across all eleven primes (Figure 2). The constant agrees with the inverse of the small- τ screening of Section 5 to $\sim 1.5\%$ ($1/D$ from the first bin gives 1.101 ± 0.003 against $k_0 = 1.083 \pm 0.007$); the full closure algebra across τ — where k_0 stays flat while D falls — involves the DW and pipeline factors and is folded into Open Problem 3. Note $f(1/2) = 0$: the vanishing of midpoint spots at the fold is the sampling Nyquist condition.

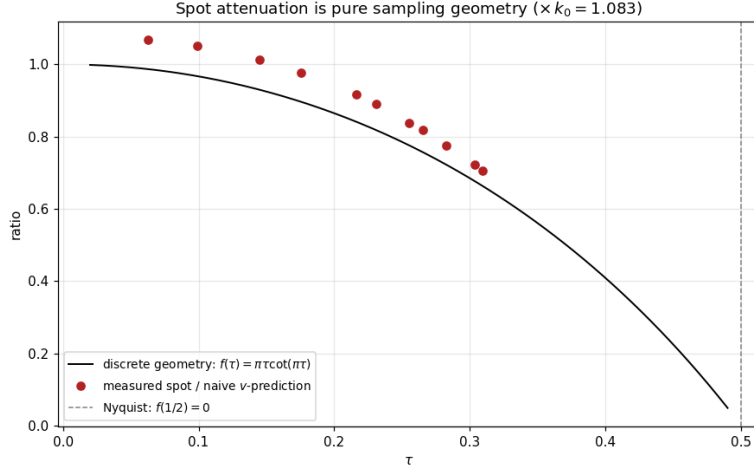


FIGURE 2. Spot attenuation against the naive v -based prediction collapses onto the discrete sampling geometry $\pi\tau \cot(\pi\tau)$, times the flat constant $k_0 = 1.083 \pm 0.007$.

4. THE RULER IDENTITY AND THE ORIGIN OF THE DRESSING

The Gram coupling between any two wave columns on the real grid is an exact functional of $\hat{G}$:

$$\langle \cos(\omega_a t) \cos(\omega_b t) \rangle = \frac{1}{2} \operatorname{Re} \left[\hat{G}(\omega_a - \omega_b) + \hat{G}(\omega_a + \omega_b) \right],$$

(and the analogous sine combinations), verified numerically to 5×10^{-8} against directly computed Grams. The omitted-variable transfer that biased restricted-basis channel measurements [6, 7] is therefore entirely a property of the diffraction pattern. Decomposing the transfer line by line shows it is *collective* — all eleven omitted small-prime lines contribute with the same sign, none dominating — and the carrier frequencies sit not on arithmetic lines but on the *thermal diffuse scattering*: the rising flank of $|\hat{G}|$ between the hyperuniform dark field and the Bragg comb, at 4–18 times the local dark floor, of the magnitude a one-dimensional warm crystal predicts ($\sqrt{(1 - e^{-\omega^2 \sigma^2})/2n} \approx 3.5 \times 10^{-3}$ at $\omega = 6.5$ against measured $5\text{--}9 \times 10^{-3}$). In one sentence: *the bending of the regressions is consistent, in location and magnitude, with the thermal diffuse scattering of the zero crystal.*

Binned over frequency, the pattern makes contact with the Montgomery form factor [3]: with $F = n|\hat{G}|^2$ and $\alpha = \omega/L$, the sub-first-line region carries $\bar{F} = 0.002$ against the ramp value 0.036 (a factor 18 dark), while the first line-bearing bin recovers the ramp from the spots alone (0.102 vs 0.089), with the off-line continuum 10–20 times below throughout — Berry’s semiclassical construction of the GUE ramp out of prime lines [1], observed directly on the midpoint grid. At larger α the binned means fall behind the ramp in the same region where the v -channel saturates; we record, as a theory target, that the ramp theorem plus the spot law constitute a sum rule constraining $v(\tau)$ from first principles.

5. THE GAP-CHANNEL SUM RULE AND THE SCREENING FUNCTION

Carrying the derived U_q through the difference geometry of the gap channel gives a closed-form prediction,

$$v(\tau) = \frac{2}{\pi} \sin(\pi\tau),$$

which is the *rigid* (unscreened) prediction. Its two constants organize the data once the screening below is included: the measured complete-basis onset is $(1.7\text{--}1.8)\tau = 2\tau D(0)$, and the measured near-fold level is $\approx 0.47 = (2/\pi) D(\text{fold})$. (The previous convention's onset 2.014τ matched the rigid 2τ only through an approximate cancellation of two systematics — basis-transfer inflation against screening — and is recorded as a coincidence.) Against the 132 complete-basis measurements the $\sin(\pi\tau)$ shape holds with free amplitude 0.530, with a systematic τ -drift of the pointwise ratio ($0.92 \rightarrow 0.74$) that is precisely the screening function below (Figure 3).

Injection calibration — adding a synthetic wave of known amplitude to the *real* zero sequence and passing it through the identical pipeline — shows the cosine (midpoint) geometry accurate to 1–9% across $\tau = 0.10\text{--}0.40$, linear in the injected amplitude (a fivefold amplitude change moves the calibration factors by $< 0.7\%$), and the pipeline nearly faithful on the sine side ($a_v = 1.00\text{--}0.85$ from $\tau = 0.06$ to 0.50). The residual,

$$D(\tau) = \frac{v_{\text{meas}}}{2\tau a_v(\tau)} = 0.91 \rightarrow 0.48 \quad (\tau : 0.05 \rightarrow 0.55),$$

is therefore a physical *screening function*: the gap (difference) channel responds to the coherent prime drive less than rigid displacement predicts, while the position (average) channel responds in full. An additive core — fluctuations independent of the lines — would give $D \equiv 1$; the measurement excludes it. The zeros move partly *collectively* under the prime waves: translations are free, compressions are resisted.

Λ -weighting and $1/k$. For prime powers the derived $v(p^k) = \frac{2}{\pi} \sin(\pi\tau) D/k$ predicts that $k v(p^k)$ lands on the prime curve. Measured (ratio test, independent of D 's shape): 1.013 ± 0.009 ($q = 4$), 1.005 ± 0.014 (8), 1.005 ± 0.009 (9), 0.953 ± 0.021 (16), 0.984 ± 0.012 (25), 0.953 ± 0.019 (27); the alternative ($\log q$ weighting, ratio $\approx k$) is excluded. The v -law is a law of Λ .

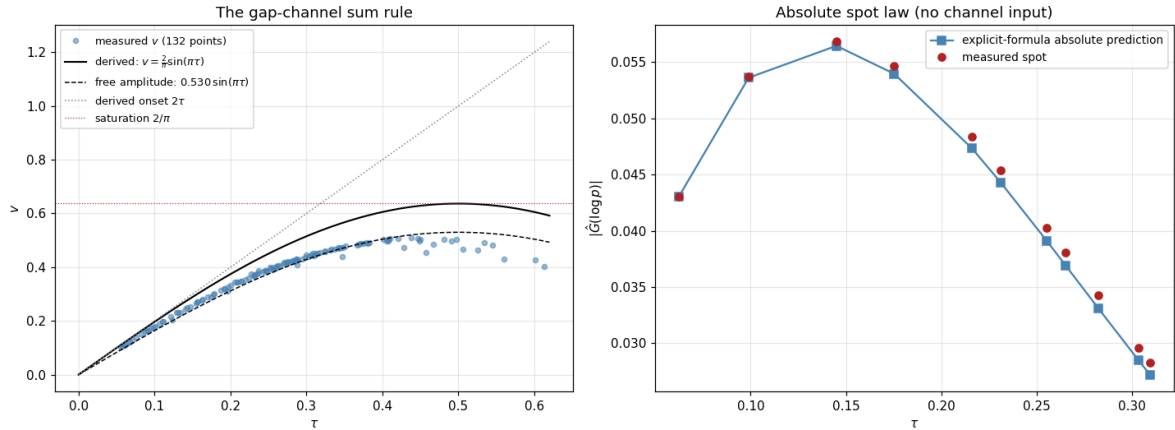


FIGURE 3. Left: the derived gap-channel sum rule against the 132 complete-basis v measurements. Right: the absolute spot law.

6. THE TWO-BRANCH RESPONSE

Is the screening a property of the gas or of the drive? Define the conditional gap response at wavevector $\kappa = 2\pi\tau$: the regression of gap-content on position-content, normalized so that a rigid displacement wave gives 1. Three measurements answer (Figure 4).

The CUE reference. For the circular unitary ensemble the response is computed as a conditional (regression) functional — the covariance of gap content on position content over 3000 free samples; no perturbed simulation is required, and equality with a potential response is not assumed — with the Diaconis–Shahshahani moments [2] ($\langle |\text{Tr } U^m|^2 \rangle = \min(m, N)$) reproduced as an internal check. CUE does *not* screen: $D_{\text{CUE}} = 1.00 \rightarrow 1.18$ from $\kappa \rightarrow 0$ to the Nyquist point.

The spontaneous branch. The same conditional response measured on the zeta grid’s *own* off-line fluctuation modes (pseudo-ensembles of 630–950 line-free frequencies per band; local unfolding, cubic detrending, and phase-scrambled surrogate floors all verified) gives:

$$D_{\text{spont}} = 0.04 (\tau=0.04), 0.24 (0.06), 0.49 (0.08), 0.66 (0.10), 0.93 (0.15), 0.96 (0.20),$$

(each point mixes a band of width $\pm 0.02L$; band-mixing biases the frozen values *upward*, so the freezing is if anything deeper, with a band systematic of order 10% on individual points; the first-line threshold spans $\tau_2 = 0.056$ – 0.070 across windows) then 1.05, 1.11, 1.18 at $\tau = 0.3, 0.4, 0.5$ — within 3% of CUE, reaching 0.34% at the Nyquist point (the $\tau = 0.3$ – 0.4 deviations are same-signed, below CUE; window-scatter error bars are pending). Two facts stand out. First, for $\tau \gtrsim 0.3$ the zero gas responds to its own noise like CUE at the few-percent level: an agreement test that, involving consecutive-gap contents rather than two-point counts alone, is not reducible to the pair-correlation function. Second, below the first prime line ($\tau_2 = \log 2/L \approx 0.067$) the gas is *frozen*: position modulations exist but gaps do not breathe — the sub-ramp (quasi-random) regime of the form factor, where the dark field is a factor 18 below the GUE ramp, is rigid, and the thaw to CUE behavior occurs across precisely the band where the first prime lines turn on.

The prime branch. The coherent arithmetic drive behaves oppositely: D_{prime} *starts* near unity at long wavelengths (where the spontaneous branch is frozen) and is screened to 0.48 at the fold (where the spontaneous branch exceeds unity). The two branches cross at $\tau^* = 0.140$ (range scan 0.13–0.15; the two branches are measured by different estimators — log-gap regression versus direct Fourier covariance — and the cross-estimator systematic is not yet budgeted). The prime waves act as quasi-static deformations of the equilibrium frame — followed adiabatically at long wavelength, resisted at short — while the spontaneous long-wave modes are collective translations that carry no compression. A theory of either branch, and of their crossing scale, is open.

7. THE POWER DEFICIT: TWO FAILED MECHANISMS AND A CONSTRAINT SET

Prime-power lines carry slightly less than full Λ weight, a deficit growing with q and k , seen consistently in three independent objects (unconditioned channel contents, diffraction spots, and mildly in the gap-ratio test): spot ratios 0.99 ($q=4$), 0.96 (8), 0.98 (9), then 0.87, 0.89, 0.81, 0.65, 0.72 for $q = 16, 25, 27, 32, 49$.

Two mechanisms were built and killed. A noiseless laboratory lattice obeying the implicit zero condition $\bar{\rho}u = -S(t+u)$ with exact explicit-formula lines reproduces a power-selective deficit of the right ordering — but necessarily rotates the power-spot phases by 10–30°; the real spots sit at 180° within 0.01–0.65° — on the strong spots ($q = 4, 8, 9$; statistical

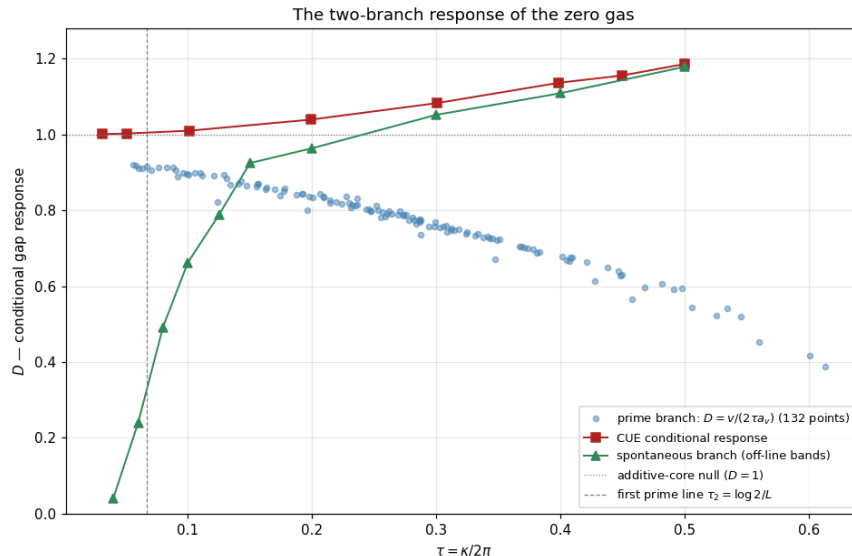


FIGURE 4. The two-branch conditional gap response: the coherent prime branch (screened), the spontaneous branch (frozen below the first prime line, CUE-like above $\tau \approx 0.3$), and the CUE reference.

uncertainty $3\text{--}5^\circ$) this comfortably excludes the predicted $11\text{--}18^\circ$ rotations — so the coherent second-order mechanism is excluded (the real field’s random component decoheres it). A Bessel-type harmonic-sampling mechanism (power spots fed by the parent prime line through the sampling exponential) preserves phase but fails quantitatively (J_4 far too small for $q = 16$; the $q = 49$ vs $q = 25$ ordering inverted relative to the parent-line hierarchy).

What survives is a sharp constraint set for the true mechanism: it is *phase-preserving* (a real multiplicative factor), *selects prime powers* while leaving primes exact at the 10^{-3} level, and *grows with both q and k* . The natural remaining candidate — second-order corrections to the $S(t)$ line amplitudes themselves — is stated as an open analytic problem.

8. SYNTHESIS AND OPEN PROBLEMS

The zero lattice, viewed through its own diffraction pattern, is a warm one-dimensional crystal: arithmetic spots whose brightnesses the explicit formula predicts absolutely; a Bragg comb whose brightness is the window thermometer; thermal diffuse scattering that is precisely the agent bending every restricted regression; and a hyperuniform dark field whose binned average rebuilds the Montgomery ramp out of prime lines. On this substrate the gap channel obeys a derived sum rule (2τ onset, $2/\pi$ scale) filtered through a measured screening function, and the conditional response splits into two branches — frozen-to-CUE for the gas’s own noise, near-unity-to-screened for the coherent primes — crossing at $\tau^* \approx 0.14$.

Open problems: (1) the power-deficit mechanism under the three constraints of Section 7; (2) a theory of the screening function $D(\tau)$ and of the spontaneous branch’s freeze-to-CUE transition, including the crossing scale $\tau^* \approx 0.14$; (3) the ramp-plus-spot-law sum rule as a first-principles derivation of the v -normalization and of k_0 ; (4) the 6% non-Edgeworth remainder of the comb and the sub-Gaussian refinement of the spot DW factor; (5) the low- τ frozen regime as a compressibility statement — what, precisely, is rigid below the first prime

line?; (6) the same diffraction program for Dirichlet L -functions: which spots move, which constants persist?

REPRODUCIBILITY

All measurements are scripts 79–95 in <https://github.com/azizran/deney> (directory `qm_riemann/`), runnable from the stored datasets; the dated research log records the two failed mechanisms of Section 7 alongside the results, together with the audit trail of the companion revisions.

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